Math 220A HW 2

Zih-Yu Hsieh

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1 D

Problem 1

Prove that all finite p-groups are solvable.

Solution: Given any finite p-group G (where p is prime), the order $|G| = p^k$ for some $k \in \mathbb{N} \setminus \{0\}$. Hence, we'll perform induction on the exponent k.

For base case $k = 1$, if $|G| = p$, then since p is prime then $G \cong \mathbb{Z}/p\mathbb{Z}$ as group (must be cyclic, hence abelian). Then, since abelian groups are vacuously solvable, G is solvable.

Now, suppose for given $n \in \mathbb{N}$, any p-group G with $|G| = p^k$ (where $k < n$) is solvable. Then, for the case of any p-group H with $|H| = p^n$, we'll consider the following statements:

1. **H has nontrivial Center:**
   If consider the conjugation action of H on itself, then all the singletons among the conjugation classes are precisely the elements in the center (since $\{h\} \subseteq H$ is a conjugation class $\iff ghg^{-1} = h$ for all $g \in H \iff gh = hg$ for all $g \in H$, which is characterized as $h \in Z(H)$, the center of H). Hhen based on the class equation, we get the following:

$$|H| = |Z(H)| + \sum_{j \in J} |\mathrm{Conj}(h_j)| \tag{1.1}$$

   Where the list of J runs through distinct $h_j \in H$, each represent a distinct conjugation class that is not singleton (and $\mathrm{Conj}(h_j)$ denotes the conuation class of h_j). Then, since by Orbit Stabilizer Theorem, one has $|H| = p^n$ being divisible by $|\mathrm{Conj}(h_j)|$ (which is not 1), then one must have $|\mathrm{Conj}(h_j)| = p^{i_j}$ for some integer $i_j > 0$, so $|\mathrm{Conj}(h_j)|$ is divisible by p. Hence, every element in the sum $\sum_{j \in J} |\mathrm{Conj}(h_j)|$ are divisible by p, $\sum_{j \in J} |\mathrm{Conj}(h_j)| = p \cdot k$ for some integer $k \in \mathbb{Z}$. Together with $|H| = p^n$, one has the following:

$$|Z(H)| = |H| - \sum_{j \in J} |\mathrm{Conj}(h_j)| = p^n - p \cdot k = p \cdot (p^{n-1} - k) \tag{1.2}$$

   This shows that $|Z(H)|$ is divisible by p, hence $|Z(H)| \neq 1$, showing the center is nontrivial.

2. **Preimage of Group Homomorphism preserves Normality:**
   Given a group homomorphism $f : K \twoheadrightarrow L$, let $N \leq M \leq L$ be subgroups such that $N \trianglelefteq M$. If consider their preimage $N' = f^{-1}(N)$ and $M' = f^{-1}(M)$ (where $N' \leq M'$), one has $f(N') \leq N$ and $f(M') \leq M$. Now, given any $r \in N'$ and $q \in M'$, if consider qrq^{-1}, we have the following:

$$f(qrq^{-1}) = f(q)f(r)f(q)^{-1} \in N \tag{1.3}$$

Since $f(q) \in M$ and $f(r) \in N$, based on normality $N \trianglelefteq M$ one has $f(q)f(r)f(q)^{-1} \in N$. Therefore, $qrq^{-1} \in f^{-1}(N) = N'$, showing that $N' \trianglelefteq M'$.

As a consequence of **1**, with $|Z(H)|$ divides $|H| = p^n$, one must have $|Z(H)| = p^k$ for some integer $0 < k \leq n$; together with $Z(H)$ being abelian, utilizing Cauchy's Theorem, there exists element $l \in Z(H)$ such that $|l| = p$, so $\langle l \rangle \subseteq H$ is a central (hence normal) subgroup of H with order p. Then, $H/\langle p \rangle$ is a quotient group with $|H/\langle l \rangle| = \frac{p^k}{p} = p^{n-1}$ also being a p-group (or a trivial group, if $n = 1$). Hence, with $(n-1) < n$, by induction $H/\langle l \rangle$ is solvable. Hence, there exists chain/tower of subgroups $\{e\} = K_0 < K_1 < \ldots < K_m = H/\langle l \rangle$, such that for each index i, $K_i \trianglelefteq K_{i+1}$, and K_{i+1}/K_i is abelian.

Now, using the condition given by **2**, pullback the tower using the projection $\pi : H \twoheadrightarrow H/\langle l \rangle$, let $K_i' := \pi^{-1}(K_i)$, it admits the following tower:

$$\{e\} < \langle l \rangle = K_0' < K_1' < \ldots < K_m' = H \tag{1.4}$$

Since π is surjective, then $K_m' = \pi^{-1}(K_m) = \pi^{-1}(H/\langle l \rangle)$; also, since $\pi^{-1}(K_0) = \pi^{-1}(\{1\}) = \ker(\pi)$, with π being projection, then $K_0' = \pi^{-1}(\{1\}) = \ker(\pi) = \langle l \rangle$. And, the reason this is a tower, is because $K_i \trianglelefteq K_{i+1}$ implies $K_i' \trianglelefteq K_{i+1}'$ by consequence of **2**.

Finally, to prove that such tower is abelian, for each index i, the projection restricts to $\pi : K_i' \twoheadrightarrow K_i$ (since π is surjective, so $\pi(K_i') = \pi(\pi^{-1}(K_i)) = K_i$). Now, given the restriction $\pi : K_{i+1}' \twoheadrightarrow K_{i+1}$, since $K_i \trianglelefteq K_{i+1}$, and K_{i+1}/K_i is abelian by assumption, then composing with the projection $\rho : K_{i+1} \twoheadrightarrow K_{i+1}/K_i$, one has $\rho \circ \pi : K_{i+1}' \twoheadrightarrow K_{i+1}/K_i$ being an surjective group homomorphism with abelian image, showing that $K_{i+1}'/\ker(\rho \circ \pi) \cong K_{i+1}/K_i$ being abelian also.

Now, notice that $\ker(\rho \circ \pi) = K_i'$: Given any $g \in K_{i+1}'$, one has $g \in \ker(\rho \circ \pi) \iff \rho(\pi(g)) = 1 \in K_{i+1}/K_i \iff \pi(g) \in K_i \iff g \in \pi^{-1}(K_i) = K_i'$. Hence, we in fact has K_{i+1}'/K_i' also being abelian, concluding that the following tower is abelian:

$$\{e\} < K_0' < K_1' < \ldots < K_m' = H \tag{1.5}$$

(Note: the reason why $\{e\} < K_0'$ is part of the abelian tower, is simply because $K_0' = \langle l \rangle$ is central, hence automatically abelian).

Hence, H (with $|H| = p^n$) is solvable. This finishes the induction.

2 D (Broken Down to Problem 2, 3)

Problem 2

> Lang Algebra Chapter 1 #49:
>
> **A group as an automorphism group.** Let G be a gruop and let $\mathsf{Set}(G)$ be the category of G-sets (i.e. sets with a G-operation / G-action). Let $F : \mathsf{Set}(G) \to \mathsf{Set}$ be the forgetful functor, which to each G-set assigns the set itself. Show that $\mathrm{Aut}(F)$ is naturally isomorphic to G.

Solution: Given that $\mathrm{Aut}(F)$ denotes the set of all natural isomorphisms from functor F to itself, to show it can be naturally isomorphic to G, we'll first claim some statements. As a side note, here $G \in \mathsf{Set}(G)$ is identified with its natural left action on itself (i.e. left multiplication), if any other action of G on itself shows up, it'll be specified.

1. **Any G-set X satisfies $\mathrm{Hom}_{\mathsf{Set}(G)}(G, X) \cong X$ as sets:**
   First, for any $f \in \mathrm{Hom}_{\mathsf{Set}(G)}(G, X)$ (which preserves left G-action), it's uniquely identified by $f(e) \in X$. Since given any $g \in G$, one has $f(g) = f(g \cdot e) = g \cdot f(e)$, so knowing $f(e) \in X$ automatically determines the whole map f.
   Conversely, for any $x \in X$, one can construct such map $f_x \in \mathrm{Hom}_{\mathsf{Set}(G)}(G, X)$ such that $f_x(e) = x$, since define $f_x : G \to X$ by $f_x(g) := g \cdot x$ for all $g \in G$, satisfies $f_x(e) = e \cdot x = x$, and $g \cdot f_x(e) = g \cdot x = f_{x(g)} = f_x(g \cdot e)$, then f_x is a G-equivariant map with $f_x(e) = x$, showing that $f_x \in \mathrm{Hom}_{\mathsf{Set}(G)}(G, X)$.
   Hence, consider the map $\mathrm{Hom}_{\mathsf{Set}(G)}(G, X) \to X$ by $f \mapsto f(e)$, this defines a natural bijection (since each $x \in X$ is endowed with a unique map $f_x \in \mathrm{Hom}_{\mathsf{Set}(G)}(G, X)$ such that $f_x(e) = x$).

2. **Natural Isomorphism on F satisfies $\mathrm{Aut}(F) \cong G$ as sets:**
   Let $\eta : F \rightsquigarrow F$ be a natural isomorphism (which, its inverse natural transformation $\eta^{-1} : F \rightsquigarrow F$ satisfies the property that for al G-set $X \in \mathsf{Set}(G)$, $\eta_X : F(X) \rightsquigarrow F(X)$ is invertible, and $(\eta_X)^{-1} = \eta_X^{-1}$).
   Now, since F is just forgetful functor, as set $F(G) = G$, and for any $X \in \mathsf{Set}(G)$, all $f \in \mathrm{Hom}_{\mathsf{Set}(G)}(G, X)$ has $Ff = f \in \mathrm{Hom}_{\mathsf{Set}}(G, X)$ (which simply forgets the G-equivariance structure, as set maps they're the same). Notice that given any $X \in \mathsf{Set}(G)$, one has $\eta_X : F(X) \rightsquigarrow F(X)$ (or $\eta_X : X \rightsquigarrow X$ as set map) being completely determined by $\eta_G(e)$ (where $\eta_G : F(G) \rightsquigarrow F(G)$, or $\eta_G : G \rightsquigarrow G$ as set maps).

   First, we'll show that $\eta_G(e) \in G$ completely determines the whole natural isomorphism: Given any $X \in \mathsf{Set}(G)$ and any $x \in X$, recall that from **1** there exists a unique G-equivariant map $f_x \in \mathrm{Hom}_{\mathsf{Set}(G)}(G, X)$ such that $f_x(e) = x$. Which, η must satisfy the following:

$$\begin{tikzcd} G \arrow[r, "f_x"] \arrow[d, "\eta_G"'] & X \arrow[d, "\eta_X"] \\ G \arrow[r, "f_x"'] & X \end{tikzcd}$$

   Hence, we get thefollowing relation (given that $\eta_G(e) \in G$):

$$\eta_X(x) = \eta_X(f_x(e)) = f_x(\eta_G(e)) = f_x(\eta_G(e) \cdot e) = \eta_G(e) \cdot f_x(e) = \eta_G(e) \cdot x \tag{2.1}$$

   Hence, given a natural isomorphism $\eta : F \rightsquigarrow F$, it's completely determined by its assignment of e, or what $e \mapsto \eta_G(e)$ is.

Next, given any $g \in G$, there exists such natural isomorphism $\rho : F \rightsquigarrow F$ such that $\rho_G(e) = g$: Given any $X \in \mathsf{Set}(G)$, define the corresponding morphism $\rho_X : X \rightsquigarrow X$ as $\eta_X(x) = g \cdot x$. Since the G-action is always invertible, it's clear that ρ_X must be set isomorphism. So, the collection $\rho = \{\rho_X : X \rightsquigarrow X \mid X \in \mathsf{Set}(G)\}$ has an „inverse" collection (since each morphism has an inverse). So, to prove it's a natural isomorphism on F, it suffices to prove that ρ is a natural transformation (since a natural transformation is an isomorphism iff every morphism is an isomorphism).

Which, for any $X, Y \in \mathsf{Set}(G)$, given any $f \in \mathrm{Hom}_{\mathsf{Set}(G)}(X, Y)$, any $x \in X$ and $h \in G$ satisfies $f(h \cdot x) = h \cdot f(x) \in Y$. Then, the collection of isomorphism ρ does the following job:

$$\rho_Y(f(x)) = g \cdot f(x) = f(g \cdot x) = f(\rho_X(x)) \tag{2.2}$$

Hence, this demonstrates that $\rho_Y \circ f = f \circ \rho_X$. So, the following diagram commutes for any such choice of $X, Y \in \mathsf{Set}(G)$, and G-equivariant map $f \in \mathrm{Hom}_{\mathsf{Set}(G)}(X, Y)$:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\rho_X"'] & Y \arrow[d, "\rho_Y"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

This shows that ρ is in fact a natural transformation on F the forgetful functor, and with every corresponding morphism being an isomorphism, it's a natural isomorphism.

Therefore, the assignment $\varphi : \mathrm{Aut}(F) \to G$ by $\eta \mapsto \eta_G(e)$ defines a bijection (since each $g \in G$ has a unique $\eta \in \mathrm{Aut}(F)$ such that $\eta_G(e) = g$).

With the map $\varphi : \mathrm{Aut}(F) \rightsquigarrow G$ being a bijection, to prove that $\mathrm{Aut}(F) \cong G$ as groups, it suffices to show that φ is indeed a group homomorphism. However, given any $\rho, \eta \in \mathrm{Aut}(F)$, we have $(\rho \circ \eta)_G = \rho_G \circ \eta_G$. Let $g = \rho_G(e)$ and $h = \eta_G(e)$, the relation proven in **2** shows that for all $k \in G$, one has $\rho_G(k) = \rho_G(e) \cdot k = gk$ (recall that here G is using left action on itself). Hence, the following relation is true:

$$(\rho \circ \eta)_G(e) = \rho_G \circ \eta_G(e) = \rho_{G(h)} = gh \tag{2.3}$$

Hence, since $\varphi(\rho) = \rho_G(e) = g$ and $\varphi(\eta) = \eta_G(e) = h$, we have $\varphi(\rho \circ \eta) = (\rho \circ \eta)_G(e) = gh = \varphi(\rho)\varphi(\eta)$. So, φ is indeed a group homomorphism, hence it's a group isomorphism between $\mathrm{Aut}(F)$ and G. So, $\mathrm{Aut}(F)$ and G are indeed isomorphic.

Problem 3

Lang Algebra Chapter 1 #53:

Let H, G, G' be groups, and let $f : H \to G$, $g : H \to G'$ be two homomorphisms. Define the notion of coproduct of these two homomorphisms over H, and show that it exists.

Solution: Here, we'll first define the *Coslice Category* over H (denoted as Grp^H). Here, define $\mathrm{Ob}(\mathsf{Grp}^H)$ to be the collection of all pair $(K, h_K : H \to K)$, where K is a group and h_K is a group homomorphism. Also, given any groups K, L with $h_K : H \to K$ and $h_L : H \to L$, define the set of morphisms $\mathrm{Hom}_{\mathsf{Grp}^H}(h_K, h_L) := \{f : K \to L \mid h_L = f \circ h_K\}$, which are all the group homomorphism $f : K \to L$ such that the following diagram commutes:

$$\begin{tikzcd}
& H \arrow[dl, "h_K"'] \arrow[dr, "h_L"] & \\
K \arrow[rr, "f"'] & & L
\end{tikzcd}$$

To show that such definitions form a category, it suffices to check that composition of morphisms is well-defined, and the endomorphism set contains identity (since associativity naturally follows from the associativity of set function composition).

First, given groups K, L, P and h_K, h_L, h_P three morphisms in Grp^H, if $f \in \mathrm{Hom}_{\mathsf{Grp}^H}(h_K, h_L)$ and $g \in \mathrm{Hom}_{\mathsf{Grp}^H}(h_L, h_P)$, the following diagram commutes:

$$\begin{tikzcd}
& H \arrow[dl, "h_K"'] \arrow[d, "h_L"'] \arrow[dr, "h_P"] & \\
K \arrow[r, "f"'] & L \arrow[r, "g"'] & P
\end{tikzcd}$$

This shows that $g \circ f$ is a morphism from h_K to h_P, or $g \circ f \in \mathrm{Hom}_{\mathsf{Grp}^H}(h_K, h_P)$, hence composition is well-defined.

Finally, extending from the above notations, the identity $\mathrm{id}_L : L \rightsquigarrow L$ satisfies the following:

$$\begin{tikzcd}
& H \arrow[dl, "h_K"'] \arrow[d, "h_L"'] \arrow[dr, "h_L"] & \\
K \arrow[r, "f"'] & L \arrow[r, "\mathrm{id}_L"'] & L
\end{tikzcd}
\qquad
\begin{tikzcd}
& H \arrow[dl, "h_L"'] \arrow[d, "h_L"'] \arrow[dr, "h_P"] & \\
L \arrow[r, "\mathrm{id}_L"'] & L \arrow[r, "g"'] & P
\end{tikzcd}$$

Hence, showing that $\mathrm{id}_L \in \mathrm{End}_{\mathsf{Grp}^H}(h_L)$, together with $\mathrm{id}_L \circ f = f$ and $g \circ \mathrm{id}_L = g$ (given $f \in \mathrm{Hom}_{\mathsf{Grp}^H}(h_K, h_L)$ and $g \in \mathrm{Hom}_{\mathsf{Grp}^H}(h_L, h_P)$). Hence, identity is well-defined, showing it's indeed a category.

To define a notion of coproduct of $f : H \to G$ and $g :\to G'$ given in the problem, it'll be a group C with group homomorphism $h_C : H \to C$, together with two morphisms $i_f \in \mathrm{Hom}_{\mathsf{Grp}^H}(f, h_C)$ and $i_g \in \mathrm{Hom}_{\mathsf{Grp}^H}(g, h_C)$, such that $i_f \circ f = i_g \circ g$, and it is initial with this property.

Which means, given any other $h_D : H \to D$ in the category that's paired up with morphisms $d_f \in \mathrm{Hom}_{\mathsf{Grp}^H}(f, h_D)$ and $d_g \in \mathrm{Hom}_{\mathsf{Grp}^H}(g, h_D)$, there exists a unique morphism $d : C \to D$, such that $d_f = d \circ i_f$ and $d_g = d \circ i_g$.

Which, by dsescription one must have $h_C = i_f \circ f = i_g \circ g$, and $h_D = d_f \circ f = d_g \circ g$, so one can implicitly assume their existence in the diagram. Which, it becomes the following:

$$\begin{tikzcd}
H \arrow[r, "f"] \arrow[d, "g"'] & G \arrow[d, "i_f"] \arrow[ddr, bend left, "d_f"] & \\
G' \arrow[r, "i_g"'] \arrow[drr, bend right, "d_g"'] & C \arrow[dr, dotted, "\exists! d"] & \\
& & D
\end{tikzcd}$$

Hence, the object C together with morphism $i_f : G \to C$ and $i_g : G' \to C$ (which are groups together with two group homomorphisms into it) can equivalently be interpreted as fibre coproduct of $f : H \to G$ and $g : H \to G'$.

The last step is to show the existence. Since in Grp one knows the existence of coproduct, a free product of the two groups together with canonical injection. So, given $f : H \to G$ and $g : H \to$

G',consider the free product $G * G'$ and the two canonical injections $i_f : G \to G * G'$ and $i_g : G' \to G * G'$, we have two group homomorphisms $i_f \circ f, i_g \circ g : H \to G * G'$.

Then, one can take a coequalizer of the two group homomorphisms: Consider he set $S = \left\{ i_f \circ f(x) \cdot \left(i_g \circ g(x)\right)^{-1} \mid x \in H \right\} \subset G * G'$ (i.e. elements in image of $i_f \circ f$, conjugates with the corresponding image in $i_g \circ g$ based on what they get mapped from in H). Take the subgroup generated by it, say $A(S) \subseteq G * G'$, and take its normal closure $\overline{A(S)} \subseteq G * G'$ (or, the smallest normal subgroup containing $A(S)$). Consider the projection $\pi : G * G' \to G * G' / \overline{A(S)}$. So, now one has the following diagram (which the left part is not necessarily commutative):

$$\begin{tikzcd} & G \arrow[dr, "i_f"] & & \\ H \arrow[ur, "f"] \arrow[dr, "g"'] & & G * G' \arrow[r, "\pi"] & G * G'/\overline{A(S)} \\ & G' \arrow[ur, "i_g"'] & & \end{tikzcd}$$

Now, given the two group homomorphisms $\pi \circ i_f : G \to G * G' / \overline{A(S)}$ and $\pi \circ i_g : G' \to G * G' / \overline{A(S)}$, notice that given any $x \in H$, one has the following:s

$$\left(\pi \circ i_f\right) \circ f(x) = \overline{i_f \circ f(x)} = \overline{i_g \circ g(x)} = \left(\pi \circ i_g\right) \circ g(x) \in G * G' / \overline{A(S)} \tag{2.4}$$

By construction, one has $\left(i_f \circ f(x)\right) \cdot \left(i_g \circ g(x)\right)^{-1} \in \overline{A(S)}$, hence $\overline{i_f \circ f(x)} = \overline{i_g \circ g(x)}$ when taking the quotient. Hence, the relation is true, and reduces to the following diagram:

$$\begin{tikzcd} H \arrow[r, "f"] \arrow[d, "g"'] & G \arrow[d, "\pi \circ i_f"] \\ G' \arrow[r, "\pi \circ i_g"'] & G * G'/\overline{A(S)} \end{tikzcd}$$

Now, given any other group D with two group homomorphisms $d_f : G \to D$ and $d_g : G' \to D$ satisfying $d_g \circ g = d_f \circ f$, by the definition, since $G * G'$ (together with $i_f : G \to G * G'$ and $i_g : G' \to G * G'$) is a coproduct of G, G', there exists a unique group homomorphis $d : G * G' \to D$, such that $d_f = d \circ i_f$ and $d_g = d \circ i_g$. Or, the following diagram is true:

$$\begin{tikzcd} G \arrow[r, "i_f"] \arrow[dr, "d_f"'] & G * G' \arrow[d, dotted, "\exists ! d"] & G' \arrow[l, "i_g"'] \arrow[dl, "d_g"] \\ & D & \end{tikzcd}$$

Then, this indicates that $\left(d \circ i_f\right) \circ f = d_f \circ f = d_g \circ g = \left(d \circ i_g\right) \circ g$, in other words for all $x \in H$, one has $d \circ \left(i_f \circ f\right)(x) = d \circ \left(i_g \circ g\right)(x)$, showing $\left(i_f \circ f(x)\right) \cdot \left(i_g \circ g(x)\right)^{-1} \in \ker(d)$. Hence, we have $A(S) \leq \ker(d)$ (since all of its generators are), showing that its normal closure $\overline{A(S)} \leq \ker(d)$ also. The, by first isomorphism theorem, d uniquely factors through $G * G' / \overline{A(S)}$, or their exists a unique $\overline{d} : G * G' / \overline{A(S)} \to D$, such that $d = \overline{d} \circ \pi$. Which, we have that $d_f = d \circ i_f = \overline{d} \circ \left(\pi \circ i_f\right)$, and $d_g = d \circ i_g = \overline{d} \circ \left(\pi \circ i_g\right)$, which forms the following diagram:

$$\begin{tikzcd}
H \arrow[r, "f"] \arrow[d, "g"'] & G \arrow[d, "\pi\circ i_f"] \arrow[ddr, bend left, "d_f"] & \\
G' \arrow[r, "\pi\circ i_g"'] \arrow[drr, bend right, "d_g"'] & G*G'/\overline{A(S)} \arrow[dr, "\overline{d}"] & \\
 & & D
\end{tikzcd}$$

Also, the uniqueness of $\overline{d}$ can be deduced from the uniqueness of d (through the property of coproduct $G*G'$) and First Isomorphism Theorem (the property of quotient $G*G'/\left(\overline{A(S)}\right)$). Hence, $G*G'/\left(\overline{A(S)}\right)$ together with $\pi\circ i_f : G\to G*G'/\left(\overline{A(S)}\right)$ and $\pi\circ i_g : G'\to G*G'/\overline{A(S)}$ is indeed a fibre coproduct of f and g, showing the existence of „coproduct“ between morphism f, g, if consider them as an object in the Coslice Category Grp^H.

3 D

Problem 4

Construct the fibre product in a category using the notation of product and equalizer. This is a hint that all small limits are built out of small products and equalizers.

Solution: Given a category C with product and equalizer, we'll explicitly construct small fibre product over arbitrary indexed set I.

Given Z, a collection of objects $\{X_i\}_{i\in I}$ contained in $\mathrm{Ob}(\mathsf{C})$, and a collection of morphisms $\{f_i : X_i\to Z\}_{i\in I}$. Let $(P, \{\pi_i : P\to X_i\})$ denotes the product of the collection $\{X_i\}_{i\in I}$ (where $\pi_i : P\to X_i$ is the corresponding projection to each component). Then, the collection of morphisms $\{f_i\circ\pi_i : P\to Z\}_{i\in I}$ as an equalizer $(E, e : E\to Z)$ (notice it satisfies the property that given any $i, j\in I$, such that $(f_i\circ\pi_i)\circ e = \left(f_j\circ\pi_j\right)\circ e$, and for any other pair $(A, a : A\to Z)$ satisfying $(f_i\circ\pi_i)\circ a = \left(f_j\circ\pi_j\right)\circ a$ for all $i, j\in I$, there exists a unique morphism $\overline{a} : A\to E$, such that $a = e\circ\overline{a}$. Diagramatically, we get the following for any $i, j\in I$:

$$\begin{tikzcd}
E \arrow[r, "e"] & P \arrow[r, shift left, "f_i\circ\pi_i"] \arrow[r, shift right, "f_j\circ\pi_j"'] & Z \\
A \arrow[u, dotted, "\exists!\overline{a}"] \arrow[ur, "a"'] & &
\end{tikzcd}$$

Now, we claim that the collection $\left(E, \{\pi_i\circ e : E\to X_i\}_{i\in I}\right)$ in fact forms a fibre product for the collection of morphisms $\{f_i : X_i\to Z\}_{i\in I}$.

First, it's clear that for any $i, j\in I$, one has $f_i\circ(\pi_i\circ e) = f_j\circ\left(\pi_j\circ e\right)$ based on construction.

Now, given any other pair $\left(A, \{a_i : A\to X_i\}_{i\in I}\right)$ that satisfies $f_i\circ a_i = f_j\circ a_j$ for all indices $i, j\in I$, notice that there exists a unique morphism $a : A\to P$ such that each $a_i = \pi_i\circ a$, based on the property of product. Diagramatically, it's as follow for all $i, j\in I$:

$$\begin{tikzcd}
 & A \arrow[dl, "a_i"'] \arrow[dr, "a_j"] \arrow[d, dotted, "\exists! a"'] & \\
X_i & P \arrow[l, "\pi_i"] \arrow[r, "\pi_j"'] & X_j
\end{tikzcd}$$

Then, notice that since each $f_i \circ a_i = f_j \circ a_j$, with the product map it extends to $f_i \circ \pi_i \circ a = f_j \circ \pi_j \circ a$. Hence, the pair $(A, a : A \to P)$ in fact equalizes all $\{f_i \circ \pi_i : P \to Z\}_{i \in I}$, showing there must exist a unique morphism $\overline{a} : A \to E$, such that $a = e \circ \overline{a}$. diagramatically, it is as follow:

$$\begin{tikzcd}
E \arrow[r, "e"] & P \arrow[r, shift left, "f_i \circ \pi_i"] \arrow[r, shift right, "f_j \circ \pi_j"'] & Z \\
A \arrow[u, dotted, "\exists! \overline{a}"] \arrow[ur, "a"'] & &
\end{tikzcd}$$

Hence, unwrap the diagram, it is presented as follow for all $i, j \in I$:

$$\begin{tikzcd}
A \arrow[dr, "\overline{a}"] \arrow[drr, bend left, "a_j"] \arrow[ddr, bend right, "a_i"'] & & \\
 & E \arrow[r, "\pi_j \circ e"'] \arrow[d, "\pi_i \circ e"] & X_j \arrow[d, "f_j"] \\
 & X_i \arrow[r, "f_i"'] & Z
\end{tikzcd}$$

Finally, such morphism $\overline{a}$ is unique, because of $\overline{a}' : A \to E$ also satisfies such property, then notice that $e \circ \overline{a}' : A \to P$ satisfies $\pi_i \circ a = a_i = \pi_i \circ (e \circ \overline{a}')$ (if put $\overline{a}'$ in the diagram above, where $\overline{a}$ is), and since a is unique based on the property of product (when gathering the a_is together), we must have $a = e \circ \overline{a}'$. However, we also know that $\overline{a}$ is the unique morphism satisfying $a = e \circ \overline{a}$ based on the property of equalizer, so it enforces $\overline{a}' = \overline{a}$.

Hence, given any pair $\left(A, \{a_i : A \to X_i\}_{i \in I}\right)$ such that $f_i \circ a_i = f_j \circ a_j$ for all indices $i, j \in I$, one must have a unique morphism $\overline{a} : A \to E$, such that $a_i = (\pi_i \circ e) \circ \overline{a}$. This shows that $\left(E, \{\pi_i \circ e : E \to X_i\}_{i \in I}\right)$ indeed satisfies the universal property of fibre product, which also proves that (small) fibre product exists if both products and equalizers exist.

4 D

Problem 5

Show that, if $F : \mathcal{C} \to \mathcal{D}$ is a natural equivalence between two categories $\mathcal{C}$ and $\mathcal{D}$, then it is both fully faithful and essentially surjective.

Solution:

I. Imlpication:

First, suppose F is a natural equivalence between two categories $\mathcal{C}$ and $\mathcal{D}$, then there exists another functor $G : \mathcal{D} \to \mathcal{C}$, paired with two natural isomorphisms $\eta : G \circ F \rightsquigarrow \mathrm{Id}_{\mathcal{C}}$ and $\varepsilon : F \circ G \rightsquigarrow \mathrm{Id}_{\mathcal{D}}$.

To verify it's fully faithful, given any $X, Y \in \mathcal{C}$, consider the functor's associated set map $F_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ by $F_{X,Y}(f) = Ff$, the goal is to prove that this is bijective.

For injectivity, notice that we also have $G_{F(X),F(Y)} : \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y)) \to \mathrm{Hom}_{\mathcal{C}}(G \circ F(X), G \circ F(Y))$. Which, given any $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ (where $G_{F(X),F(Y)} \circ F_{X,Y}(f) = (G \circ F)f$) satisfies the following diagram with η:

$$\begin{tikzcd} G \circ F(X) \arrow[r, "(G\circ F)f"] \arrow[d, "\eta_X"'] & G \circ F(Y) \arrow[d, "\eta_Y"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

Which, since η is a natural isomorphism, both η_X, η_Y are invertible. Hence, the map $\overline{\eta}_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(G \circ F(X), G \circ F(Y)) \to \mathrm{Hom}_{\mathcal{C}}(X, Y)$ by $(G \circ F)f \mapsto (\eta_Y \circ (G \circ F)f \circ \eta_X) = f$ is actually an isomorphism. Combining all maps, we get the following $\overline{\eta}_{X,Y} \circ G_{F(X),F(Y)} \circ F_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}}(X, Y)$ satisfies the following:

$$\overline{\eta}_{X,Y} \circ G_{F(X),F(Y)} \circ F_{X,Y}(f) = \overline{\eta}_{X,Y}((G \circ F)f) = f \tag{4.1}$$

Hence, this map is precisely identity on $\mathrm{Hom}_{\mathcal{C}}(X, Y)$, showing that it's in particular injective. So, $F_{X,Y}$ is injective.

For surjectivity instead, let $X' = F(X)$, $Y' = F(Y)$, then for any $f' \in \mathrm{Hom}_{\mathcal{D}}(X', Y')$ (which is the codomain of $F_{X,Y}$), based on the natural isomorphism ε, we also have the following:

$$\begin{tikzcd} X' \arrow[r, "f'"] & Y' \\ F \circ G(X') \arrow[r, "(F\circ G)f'"'] \arrow[u, "(\varepsilon_{X'})^{-1}"] & F \circ G(Y') \arrow[u, "(\varepsilon_{Y'})^{-1}"'] \end{tikzcd}$$

Hence, define $\overline{\varepsilon}_{X',Y'} : \mathrm{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y')) \to \mathrm{Hom}_{\mathcal{D}}(X', Y')$ by $(F \circ G)f' \mapsto (\varepsilon_{Y'})^{-1} \circ ((F \circ G)f') \circ \varepsilon_{X'} = f'$, it is an isomorphism. So, having the maps $G_{X',Y'} : \mathrm{Hom}_{\mathcal{D}}(X', Y') \to \mathrm{Hom}_{\mathcal{C}}(G(X'), G(Y'))$, $F_{G(X'),G(Y')} : \mathrm{Hom}_{\mathcal{C}}(G(X'), G(Y')) \to \mathrm{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y'))$, the composition $\overline{\varepsilon}_{X',Y'} \circ F_{G(X'),G(Y')} \circ G_{X',Y'} : \mathrm{Hom}_{\mathcal{D}}(X', Y') \to \mathrm{Hom}_{\mathcal{D}}(X', Y')$ is in fact identity (since $f' \mapsto \overline{\varepsilon}(X', Y')((F \circ G)f') = f'$). So, with $\overline{\varepsilon}_{X',Y'}$ being an isomorphism, $F_{G(X'),G(Y')} \circ G_{X',Y'}$ is also an isomorphism, hence $F_{G(X'),G(Y')}$ is surjective.

Finally, since $\mathrm{Hom}_{\mathcal{C}}(G(X'), G(Y')) = \mathrm{Hom}_{\mathcal{C}}(G \circ F(X), G \circ F(Y))$ is naturally isomorphic to $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ (via η), then so is $\mathrm{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y'))$ and $\mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ (via F applies to η, since functors preserve isomorphisms). Hence, if $F_{G(X'),G(Y')}$ is surjective (which has codomain $\mathrm{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y'))$), via the natural isomorphism of η, one can also conclude that $F_{X,Y}$ is also surjective (with codomain $\mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$).

The last piece is essentially surjective: Given any $Y \in \mathcal{D}$, since we have $F \circ G(Y) \cong Y$ (due to the fact that $\varepsilon : F \circ G \Rightarrow \mathrm{Id}_{\mathcal{D}}$), showing it's essentially surjective.

5 ND (Split to Problem 6-12) (10,12 not done)

Problem 6

Definition: 1

Let R be a commutative ring. If S is any set, we denote by $L_{R(S)}$ the set of all finite linear combinations of elements in S over R – that is, the set of formal sums $\sum_i a_i \cdot s_i$ where $a_i \in R$ and $s_i \in S$, and i runs through a finite set.

Define a suitable abelian group structure on $L_{R(S)}$ and scaling of R on $L_{R(S)}$. Show that $L_{R(S)}$ is a module over R.

Solution: For this purpose, the element $0 \in L_{R(S)}$ needs to first be defined: regardless of the set $\{s_1, ..., s_n\} \subset S$, the sume $\sum_{i=1}^n 0 \cdot s_i$ will be defined to be the same, namely the $0 \in L_{R(S)}$. Which, given any list $\{s_1, ..., s_n\} \subset S$, if $s_{n+1} \notin \{s_1, ..., s_n\}$, define the expression $\sum_{i=1}^n a_i \cdot s_i + 0 \cdot s_{n+1} = \sum_{i=1}^n a_i \cdot s_i$.

With the notion of 0, given any two finite formal sums $\sum_i a_i \cdot s_i, \sum_j b_j \cdot s_j$, by adding suitable terms with coefficient 0 to both expressions, WLOG one can say the two expressions are $\sum_{i=1}^n a_i \cdot s_i$ and $\sum_{i=1}^n b_i \cdot s_i$. Which, define the addition as follow:

$$\sum_{i=1}^n a_i \cdot s_i + \sum_{i=1}^n b_i \cdot s_i := \sum_{i=1}^n (a_i + b_i) \cdot s_i \tag{5.1}$$

where each $a_i + b_i \in R$ uses the addition structure of R.

Finally, for scalar multiplication, given any $r \in R$ and $\sum_i a_i \cdot s_i \in L_{R(S)}$, define scaling of r as follow:

$$r \cdot \left(\sum_i a_i \cdot s_i\right) := \sum_i (ra_i) \cdot s_i \tag{5.2}$$

To show that $L_{R(S)}$ forms a left R-Module structure, first it is evident that any $v \in L_{R(S)}$ satisfies $v + 0 = v$ (since 0 provides no extra components, and add nothing to the existing coefficients); also, addition is associative, since given $u, v, w \in L_{R(S)}$ (WLOG, say $u = \sum_{i=1}^n u_i \cdot s_i$, $v = \sum_{i=1}^n v_i \cdot s_i$, and $w = \sum_{i=1}^n w_i \cdot s_i$), then we have the following:

$$\begin{aligned}
(u+v)+w &= \left(\sum_{i=1}^n u_i \cdot s_i + \sum_{i=1}^n v_i \cdot s_i\right) + \sum_{i=1}^n w_i \cdot s_i = \sum_{i=1}^n (u_i + v_i) \cdot s_i + \sum_{i=1}^n w_i \cdot s_i \\
&= \sum_{i=1}^n (u_i + v_i + w_i) \cdot s_i = \sum_{i=1}^n u_i \cdot s_i + \left(\sum_{i=1}^n v_i \cdot s_i + \sum_{i=1}^n w_i \cdot s_i\right) \\
&= u + (v + w)
\end{aligned} \tag{5.3}$$

Which proves the associativity.

Now, about scalar multiplication, given any $v = \sum_i a_i \cdot s_i \in L_{R(S)}$, $0, 1 \in R$ satisfy the following:

$$0 \cdot v = 0 \cdot \left(\sum_i a_i \cdot s_i \right) = \sum_i (0 \cdot a_i) \cdot s_i = 0 \tag{5.4}$$

$$1 \cdot v = 1 \cdot \left(\sum_i a_i \cdot s_i \right) = \sum_i (1 \cdot a_i) \cdot s_i = v \tag{5.5}$$

Finally, about distributivity, given any $r, t \in R$ and $u, v \in L_{R(S)}$ (say $u = \sum_i u_i \cdot s_i$ and $v = \sum_i v_i \cdot s_i$), the following two equations are true:

$$\begin{aligned}(r+t)\cdot(u+v) &= (r+t)\cdot\left(\sum_i u_i \cdot s_i + \sum_i v_i \cdot s_i\right) \\ &= (r+t)\cdot \sum_i (u_i + v_i)\cdot s_i = \sum_i (ru_i + rv_i + tu_i + tv_i)\cdot s_i \\ &= \sum_i (ru_i)\cdot s_i + \sum_i (rv_i)\cdot s_i + \sum_i (tu_i)\cdot s_i + \sum_i (tv_i)\cdot s_i \\ &= r\cdot\left(\sum_i u_i \cdot s_i\right) + r\cdot\left(\sum_i v_i \cdot s_i\right) + t\cdot\left(\sum_i u_i \cdot s_i\right) + t\cdot\left(\sum_i v_i \cdot s_i\right) \\ &= r\cdot u + r\cdot v + t\cdot u + t\cdot v\end{aligned} \tag{5.6}$$

$$\begin{aligned}(rt)\cdot v = (rt)\cdot\left(\sum_i u_i \cdot s_i\right) &= \sum_{i(rtu_i)} \cdot s_i = r\cdot\left(\sum_i (tu_i)\cdot s_i\right) \\ = r\cdot\left(t\cdot\left(\sum_i u_i\cdot s_i\right)\right) &= r\cdot(t\cdot v)\end{aligned} \tag{5.7}$$

Hence, $L_{R(S)}$ under such addition / multiplication structure forms a left R-module.

Problem 7

Suppose that $R = \mathbb{R}$ is the ring/field of real numbers and let $S = \mathbb{N}$ denote the set of natural numbers, what is the $\mathbb{R}$-vector space $L_{\mathbb{R}}(\mathbb{N})$? Can you find a better way to write this vector space?

Solution: The vector space $L_{\mathbb{R}}(\mathbb{N})$ can also be characterized as the direct sum $\bigoplus_{n\in\mathbb{N}} \mathbb{R}$ (since there are countable generators, defined with finite sums, and in between any two generators there is no relations).

Another way of characterizing it can be thought as follow: Consider the set of functions $F = \{f : \mathbb{N} \to \mathbb{R} \mid f(n) \neq 0 \text{ for finite } n \in \mathbb{N}\}$, which because the codomain is $\mathbb{R}$ (which has a natural addition / scalar multiplication structure on itself)m then this automaticaly forms F into an $\mathbb{R}$-vector space.

Now, for each $n \in \mathbb{N}$, let $e_n : \mathbb{N} \to \mathbb{R}$ denotes the function satisfying $e_{n(m)} = \delta_{nm}$ (the kronecker delta). Then, notice that for each function $f \in F$, if $n_1, ..., n_k \in \mathbb{N}$ are the only natural numbers being evaluated to have nonzero ouput, then $f = \sum_{i=1}^k f(n_i) \cdot e_{n_i}$:

$$m \notin \{n_1, ..., n_k\} \implies \left(\sum_{i=1}^k f(n_i) \cdot e_{n_i}\right)(m) = 0 \tag{5.8}$$

$$\forall j \in \{1, \ldots, k\}, \quad \left(\sum_{i=1}^{k} f(n_i) \cdot e_{n_i}\right)(n_j) = \sum_{i=1}^{k} f(n_i) \cdot \delta_{n_i n_j} = f(n_j) \cdot 1 = f(n_j) \tag{5.9}$$

Hence, $f = \sum_{i=1}^{k} f(n_i) \cdot e_{n_i}$, showing the list $\{e_n\}_{n\in\mathbb{N}}$ in fact generates F.

Also, it is evident that each e_n has no relation with each other (which can be shown through linear independence): Given any list $e_{n_1}, \ldots, e_{n_k}$, suppose $a_1, \ldots, a_k \in \mathbb{R}$ satisfies $\sum_{i=1}^{k} a_i \cdot e_{n_i} = 0$, then for each $j \in \{1, \ldots, k\}$, the following is true:

$$0 = \left(\sum_{i=1}^{k} a_i \cdot e_{n_i}\right)(n_j) = \sum_{i=1}^{k} a_i \cdot \delta_{n_i n_j} = a_j \cdot 1 = a_j \tag{5.10}$$

Hence, each $a_j = 0$, showing that the list is linearly independent. This shows that $\{e_n\}_{n\in\mathbb{N}}$ forms a basis of the vector space F (which they're linearly independent, and their finite $\mathbb{R}$-linear combinations generate the whole F). So, this is another way of identifying $L_{\mathbb{R}}(\mathbb{N})$.

Problem 8

Definition: 2

Consider the module $L_R(M \times N)$ and define a submodule $L' \subset L_R(M \times N)$ generated by all elements of the following form:

- $r(x, y) - (rx, y)$
- $r(x, y) - (x, ry)$
- $(x_1 + x_2, y) - ((x_1, y) + (x_2, y))$
- $(x, y_1 + y_2) - ((x, y_1) + (x, y_2))$

where $r \in R$, $x, x_1, x_2 \in M$, and $y, y_1, y_2 \in N$. We then define the quotient module

$$M \otimes_R N := L_R(M \times N)/L' \tag{5.11}$$

and call it the tensor product of M and N.

Definition: 3

If P is another R-module, a bilinear map is a map $B : M \times N \to P$ that is linear with respect to both entries – that is, for all $x_0 \in M$ the map $B(x_0, \_) : N \to P$ is linear and for all $y_0 \in N$ the map $B(\_, y_0) : M \to P$ is linear.

Construct a natural bilinear map $\pi : M \times N \to M \otimes_R N$ and show that it is universal in the following sense: for any bilinear map $B : M \times N \to P$, there is a unique *linear* map $h : M \otimes_R N \to P$ such that $B = h \circ \pi$. Conclude that for any R-module P, we have a canonical identification

$$\mathrm{Hom}_R(M \times_R N, P) \cong \mathrm{Bil}(M \times N, P) \tag{5.12}$$

between the set of R-module homomorphisms from $M \otimes_R N$ to P, and the set of bilinear maps from $M \times N$ to P.

Solution:

I. The Bilinear ap π:

For simplicity, given any $m \in M$ and $n \in N$, we'll denote $m \otimes n := \overline{(m,n)} \in M \otimes_R N = L_R(M \times N)/L'$ (the quotient of (m,n) in $M \otimes_R N$). Then, given the relation of L', we get the following relation:

- $r(x \otimes y) = (rx) \otimes y$, and $r(x \otimes y) = x \otimes (ry)$.
- $(x_1 + x_2) \otimes y = x_1 \otimes y + x_2 \otimes y$.
- $x \otimes (y_1 + y_2) = x \otimes y_1 + x \otimes y_2$.

Where $r \in R$, $x, x_1, x_2 \in M$ and $y, y_1, y_2 \in N$.

Hence, define the map $\pi : M \times N \to M \otimes_R N$ by $\pi(m,n) = m \otimes n$, this map is well-defined (since each $(m,n) \in M \times N$ corresponds to a unique representative in $M \otimes_R N := L_R(M \times N)/L')$. Then, by the property of tensor products, one gets the following for any $r, t \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$:

$$\begin{aligned}\pi(rm_1 + tm_2, n) &= (rm_1 + tm_2) \otimes n = (rm_1) \otimes n + (tm_2) \otimes n \\ &= r(m_1 \otimes n) + t(m_2 \otimes n) = r \cdot \pi(m_1, n) + t \cdot \pi(m_2, n)\end{aligned} \tag{5.13}$$

$$\begin{aligned}\pi(m, rn_1 + tn_2) &= m \otimes (rn_1 + tn_2) = m \otimes (rn_1) + m \otimes (tn_2) \\ &= r(m \otimes n_1) + t(m \otimes n_2) = r \cdot \pi(m, n_1) + t \cdot \pi(m, n_2)\end{aligned} \tag{5.14}$$

The above two equations show that π is naturally bilinear.

II. Universal Property of π:

Given any R-module P together with a bilinear map $B : M \times N \to P$, suppose for the moment that there exists an R-linear map $h : M \otimes_R N \to P$ satisfies $B = h \circ \pi$, then for all $m \in M$ and $n \in N$, we have the following:

$$B(m,n) = h \circ \pi(m,n) = h(m \otimes n) \tag{5.15}$$

Now, notice that since $L_R(M \times N)$ is generated by finite formal linear combination of $(m,n) \in M \times N$, then $M \otimes_R N = L_{R(M \times N)}/L'$ is instead generated by finite linear combination of the quotient $\overline{(m,n)} = m \otimes n$. Then, since h above must satisfy $h(m \otimes n) = B(m,n)$, then h fixes where the generators of $M \otimes_R N$ goes, hence fixes the whole R-linear map h. This implies that h is uniquely determined by B.

Then, to show that h exists, it suffices to show that it's well-defined on the generators (since the definition of being an R-linear map automatically extends it to be well-defined on any finite R-linear combinations of $m \otimes n \in M \otimes_R N$). To show that any possible representatives eventually gets send to the same element via h, consider any $a, b, c, d \in R$, $m_1, m_2 \in M$, and $n_1, n_2 \in N$. Which, the following is true:

$$\begin{aligned}B(am_1 + bm_2, cn_1 + dn_2) &= (ac) \cdot B(m_1, n_1) + (ad) \cdot B(m_1, n_2) \\ &\quad + (bc) \cdot B(m_2, n_1) + (bd) \cdot B(m_2, n_2)\end{aligned} \tag{5.16}$$

The above is based on the bilinearity of B. Which, notice that on the tensor product level, the following is true:

$$\begin{aligned}(am_1 + bm_2) \otimes (cn_1 + dn_2) &= (ac) \cdot (m_1 \otimes n_1) + (ad) \cdot (m_1 \otimes n_2) \\ &\quad + (bc) \cdot (m_2 \otimes n_1) + (bd) \cdot (m_2 \otimes n_2)\end{aligned} \tag{5.17}$$

Hence, this shows the following:

$$\begin{aligned}
h((am_1 + bm_2) \otimes (cn_1 + dn_2)) &= B(am_1 + bm_2, cn_1 + dn_2) \\
&= (ac) \cdot B(m_1, n_1) + (ad) \cdot B(m_1, n_2) \\
&\quad + (bc) \cdot B(m_2, n_1) + (bd) \cdot B(m_2, n_2) \\
&= (ac) \cdot h(m_1 \otimes n_1) + (ad) \cdot h(m_1 \otimes n_2) \\
&\quad + (bc) \cdot h(m_2 \otimes n_1) + (bd) \cdot h(m_2 \otimes n_2) \\
&= h((ac) \cdot (m_1 \otimes n_1) + (ad) \cdot (m_1 \otimes n_2) \\
&\quad + (bc) \cdot (m_2 \otimes n_1) + (bd) \cdot (m_2 \otimes n_2))
\end{aligned} \tag{5.18}$$

Which, it implies that the definition $h(m \otimes n) = B(m, n)$ (together with h being an R-linear map) is well-defined (since under different representatives of the same element, h still acts the same). Hence, as an R-linear map $h : M \otimes_R N \to P$ by $h(m \otimes n) = B(m, n)$ is well-defined.

This finishes the proof about the unversal property, that every bilinear map $B : M \times N \to P$ admits a unique R-linear map $h : M \otimes_R N \to P$ with $B = h \circ \pi$.

III. Isomorphism between $\mathrm{Hom}_R(M \otimes_R N, P)$ and $\mathrm{Bil}(M \times N, P)$:

First, notice that linear combinations of R-linear maps (respectively, R-Bilinear maps) are still R-linear map (respectively, R-Bilinear map). Hence, $\mathrm{Hom}_R(M \otimes_R N, P), \mathrm{Bil}(M \times N, P)$ are both R-modules also.

Now, notice that the map $\_ \circ \pi : \mathrm{Hom}_R(M \otimes_R N, P) \to \mathrm{Bil}(M \times N, P)$ is a module homomorphism:

- First, given any R-linear map $h : M \otimes_R N \to P$, we have the following for any $r, t \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$:

$$\begin{aligned}
h \circ \pi(rm_1 + tm_2, n) &= h((rm_1 + tm_2) \otimes n) = h(r(m_1 \otimes n) + t(m_2 \otimes n)) \\
&= r \cdot h(m_1 \otimes n) + t \cdot h(m_2 \otimes n) \\
&= r \cdot h \circ \pi(m_1, n) + t \cdot h \circ \pi(m_2, n)
\end{aligned} \tag{5.19}$$

$$\begin{aligned}
h \circ \pi(m, rn_1 + tn_2) &= h(m \otimes (rn_1 + tn_2)) = h(r(m \otimes n_1) + t(m \otimes n_2)) \\
&= r \cdot h(m \otimes n_1) + t \cdot h(m \otimes n_2) \\
&= r \cdot h \circ \pi(m, n_1) + t \cdot h \circ \pi(m, n_2)
\end{aligned} \tag{5.20}$$

The two equalities show that $h \circ \pi$ is a bilinear map, henc $h \circ \pi \in \mathrm{Bil}(M \times N, P)$, the map $\_ \circ \pi$ is well-defined.

The reason it's an R-linear map, is because given any $f, g \in \mathrm{Hom}_R(M \otimes N, P)$, $a, b \in R$, and any $m \in M$ and $n \in N$, the following is true:

$$\begin{aligned}
(a \cdot f + b \cdot g) \circ \pi(m, n) &= (a \cdot f + b \cdot g)(m \otimes n) \\
&= a \cdot f(m \otimes n) + b \cdot g(m \otimes n) \\
&= a \cdot f \circ \pi(m, n) + b \cdot g \circ \pi(m, n)
\end{aligned} \tag{5.21}$$

Hence, $(a \cdot f + b \cdot g) \circ \pi = a \cdot (f \circ \pi) + b \cdot (g \circ \pi)$, showing $\_ \circ \pi$ is an R-linear map.

Finally, it's a bijection, because for any bilinear map $B \in \mathrm{Bil}(M \times N, P)$, the universal property states there exists $h \in \mathrm{Hom}_R(M \otimes N_R, P)$ such that $B = h \circ \pi$, showing $\_ \circ \pi$ is surjective; and, it's injective is because if $h, h' \in \mathrm{Hom}_R(M \otimes_R N, P)$ satisfies $B = h \circ \pi = h' \circ \pi \in \mathrm{Bil}(M \times N, P)$, since both h, h' satisfies $B = h \circ \pi = h' \circ \pi$, the universal property again states $h = h'$, hence $\_ \circ \pi$ is injective.

So, this identifies that $\mathrm{Hom}_{R(M \otimes_R N, P)} \cong \mathrm{Bil}(M \times N, P)$.

Problem 9

Is $M \otimes_R N$ the categorical pushout of M and N in the category R-Mod of R-modules? If so, prove it. If not, what is the pushout of M and N?

Solution: We'll first construct the pushout (or coproduct) of any R-module M and N, and show that in general it's different from $M \otimes_R N$ by providing a counterexample.

I. Coproduct in R-Mod:

Given arbitrary R-modules M, N, consider the direct sum $M \oplus N$ together with the two canonical injection $\iota_1 : M \to M \oplus N$ and $\iota_2 : N \to M \oplus N$. It is evident that it satisfies the first property of coproduct (which are two maps from M, N respectively into the R-module).

Then, to show its universality, given any other R-module P together with two maps $h_1 : M \to P$ and $h_2 : N \to P$, the map $h : M \oplus N \to P$ by $h(m, n) = h_1(m) + h_2(n)$ satisfies the following:

- First, given any $(m, n), (m', n') \in M \oplus N$, and $a, b \in R$, the following is true:

$$\begin{aligned} h(a(m,n) + b(m',n')) &= h(am + bm', an + bn') = h_1(am + bm') + h_2(an + b_{n'}) \\ &= ah_1(m) + bh_1(m') + ah_2(n) + bh_2(n') \\ &= a(h_1(m) + h_2(n)) + b(h_1(m') + h_2(n')) \\ &= a \cdot h(m,n) + b \cdot h(m',n') \end{aligned} \tag{5.22}$$

  Hence, h is a Module homomorphism.
- For any $m \in M$ and $n \in N$, we have the following two equality:

$$h \circ \iota_1(m) = h(m, 0) = h_1(m) + h_2(0) = h_1(m) \tag{5.23}$$

$$h \circ \iota_2(n) = h(0, n) + h_1(0) + h_2(n) = h_2(n) \tag{5.24}$$

  Hence, $h \circ \iota_1 = h_1$ and $h \circ \iota_2 = h_2$.
- h is the unique morphism with such property: Suppose given $h' : M \oplus N \to P$ also satisfies such property, then we must have $h' \circ \iota_1 = h_1$ and $h' \circ \iota_2 = h_2$. Then, for all $m \in M$ and $n \in N$, we must have $h_1(m) = h' \circ \iota_1(m) = h'(m, 0)$, and $h_2(n) = h' \circ \iota_2(n) = h'(0, n)$. So, $h'(m, n) = h'(m, 0) + h'(0, n) = h_1(m) + h_2(n)$, showing that $h' = h$.

These conditions show that $M \oplus N$ together with the two canonical injections $\iota_1 : M \to M \oplus N$ and $\iota_2 : N \to M \oplus N$, is initial with such property. Meaning given the pair $(P, h_1 : M \to P, h_2 : N \to P)$, there exists unique $h : M \oplus N \to P$, such that the following diagram commutes:

$$\begin{tikzcd} M \arrow[r, "\iota_1"] \arrow[dr, "h_1"'] & M \oplus N \arrow[d, dotted, "\exists! h"'] & N \arrow[l, "\iota_2"'] \arrow[dl, "h_2"] \\ & P & \end{tikzcd}$$

Hence, $M \oplus N$ with conanical injection ι_1, ι_2 forms the coproduct of M, N in R-Mod.

II. Counterexample to $M \otimes_R N$ being Pushout/Coproduct:

Here's a simple one: Consider $0 \otimes_{\mathbb{Z}} \mathbb{Z}$, since everything is in the form $0 \otimes n$ for some $n \in \mathbb{Z}$; then by the property of tensor product, $2 \cdot (0 \otimes n) = (n \cdot 0) \otimes n = 0 \otimes n$, hence $0 \otimes n = 0$, or $0 \otimes_{\mathbb{Z}} \mathbb{Z} = 0$. However, the actual coproduct is $0 \oplus \mathbb{Z} \cong \mathbb{Z} \neq 0$, showing that $0 \otimes_{\mathbb{Z}} \mathbb{Z}$ is actually not a coproduct in $\mathbb{Z}$-Mod.

Problem 10

Given A, B two commutative rings with ring homomorphisms $R \to A$ and $R \to B$, this makes A, B into R-modules. Let $A \otimes_R B$ denote the tensor product of A and B as R-modules.

In this setup, show that $A \otimes_R B$ is again a (commutative) R-algebra.

Solution: Here, assume we know that tensor product (of multiple R-modules) is associative (up to isomorphism), and the property that any multilinear map uniquely factors through tensor products. Then, since A, B both have their own ring structure, then consider the following R-multilinear map $Q : A \times B \times A \times B \to A \otimes_R B$:

$$Q(a, b, a', b') = (aa') \otimes (bb') \tag{5.25}$$

This is multilinear, because both ring multiplication and tensor products are bilinaer. Then, the property of tensor product implies that it unique factors through $(A \otimes_R B) \otimes_R (A \otimes_R B)$, say through an R-linear map $\overline{Q} : (A \otimes_R B) \otimes_R (A \otimes_R B) \to A \otimes_R B$ as follow:

$$\overline{Q}((a \otimes b) \otimes (a' \otimes b')) = (aa') \otimes (bb') \tag{5.26}$$

Also, if we consider the bilinear projection $\pi : (A \otimes_R B) \times (A \otimes_R B) \to (A \otimes_R B) \otimes_R (A \otimes_R B)$ by $\pi(a \otimes b, a' \otimes b') = (a \otimes b) \otimes (a' \otimes b')$, then the bilinear map $M = \overline{Q} \circ \pi : (A \otimes_R B) \times (A \otimes_R B) \to A \otimes_R B$ as follow:

$$M(a \otimes b, a' \otimes b') = \overline{Q}((a \otimes b) \otimes (a' \otimes b')) = (aa') \otimes (bb') \tag{5.27}$$

Which, M (as an R-bilinear map) can also be viewed as a multiplication on $A \otimes_R B$, since it satisfies associativity as follow:

$$\begin{aligned} M(a \otimes b, M(a' \otimes b', a'' \otimes b'')) &= M(a \otimes b, (a'a'') \otimes (b'b'')) = (a(a'a'')) \otimes (b(b'b'')) \\ &= ((aa')a'') \otimes ((bb')b'') = M((aa') \otimes (bb'), a'' \otimes b'') \\ &= M(M(a \otimes b, a' \otimes b'), a'' \otimes b'') \end{aligned} \tag{5.28}$$

Also, distributivity naturally follows from the fact that M is bilinear. Hence, this makes $A \otimes_R B$ into an R-algebra, with multiplication M.

Problem 11

Prove that $A \otimes_R B$ is the pushout (coproduct) of A and B in the category $\mathsf{R\text{-}Alg}$ of R-algebras.

Solution: First, it's clear that given $A \otimes_R B$, there are natural R-algebra homomorphisms $i_1 : A \to A \otimes_R B$ and $i_2 : B \to A \otimes_R B$ by $i_1(a) = a \otimes 1$ and $i_2(b) = 1 \otimes b$ (which since $i_1(r \cdot a) = (m_A(r)a) \otimes 1 = r \cdot (a \otimes 1) = r \cdot i_1(a)$, it's an R-module homomorphism together with the additive property; also, we have $i_1(aa') = (aa') \otimes 1 = (a \otimes 1)(a' \otimes 1) = i_1(a) \cdot i_1(a')$, which is also an R-algebra homomorphism. Similar logic applies to i_2).

Now, given any other R-algebra C, and suppose two R-algebra homomorphisms $f : A \to C$ and $g : B \to C$ are given. Then, if we consider the map $B : A \times B \to C$ by $B(a, b) = f(a)g(b)$, notice that this map is R-bilinear. Given any $r, t \in R$, $a, a' \in A$, and $b, b' \in B$, we have the following:

$$\begin{aligned} B(r \cdot a + t \cdot a', b) &= f(r \cdot a + t \cdot a')g(b) = (r \cdot f(a) + t \cdot f(a'))g(b) \\ &= r \cdot (f(a)g(b)) + t \cdot (f(a')g(b)) = r \cdot B(a,b) + t \cdot B(a',b) \end{aligned} \tag{5.29}$$

$$\begin{aligned} B(a, r \cdot b + t \cdot b') &= f(a)g(r \cdot b + t \cdot b') = f(a)(r \cdot g(b) + t \cdot g(b')) \\ &= r \cdot (f(a)g(b)) + t \cdot (f(a)g(b')) = r \cdot B(a,b) + t \cdot B(a,b') \end{aligned} \tag{5.30}$$

This shows that B is R-bilinear, hence it uniquely factors through the tensor product, say by the R-module homomorphism $h : A \otimes_R B \to C$ by $h(a \otimes b) = f(a)g(b)$.

However, this also turns h into a ring homomorphism, since we have the following:

$$\begin{aligned} h((a \otimes b) \cdot (a' \otimes b')) &= h((aa') \otimes (bb')) = f(aa')g(bb') = f(a)f(a')g(b)g(b') \\ &= (f(a)g(b)) \cdot (f(a')g(b')) = h(a \otimes b) \cdot h(a' \otimes b') \end{aligned} \tag{5.31}$$

Since it's both an R-module homomorphism and a commutative ring homomorphism, h is in fact an R-algebra homomorphism. And, it satisfies the following, for all $a \in A$ and $b \in B$:

$$f(a) = f(a)g(1) = h(a \otimes 1) = h \circ i_1(a) \tag{5.32}$$

$$g(b) = f(1)g(b) = h(1 \otimes b) = h \circ i_2(b) \tag{5.33}$$

Hence, $f = h \circ i_1$ and $g = h \circ i_2$.

Finally, to show uniqueness of such map h, suppose some R-algebra homomorphism $h' : A \otimes_R B \to C$ satisfies $f = h' \circ i_1$ and $g = h' \circ i_2$ also, then for all $a \in A$ and $b \in B$, we have the following:

$$\begin{aligned} h'(a \otimes b) &= h'((a \otimes 1)(1 \otimes b)) = h'(a \otimes 1) \cdot h'(1 \otimes b) \\ &= (h' \circ i_1(a)) \cdot (h' \circ i_2(b)) = f(a)g(b) = h(a \otimes b) \end{aligned} \tag{5.34}$$

This shows that $h' = h$, proving the uniqueness. Hence, $A \otimes_R B$, $i_1 : A \to A \otimes_R B$, and $i_2 : B \to A \otimes_R B$ satisfies the following commutative diagram (given some other R-algebra C with $f : A \to C$ and $g : B \to C$ being R-algebra homomorphisms):

$$\begin{tikzcd} A \arrow[r, "i_1"] \arrow[dr, "f"'] & A \otimes_R B \arrow[d, dashed, "\exists! h"] & B \arrow[l, "i_2"'] \arrow[dl, "g"] \\ & C & \end{tikzcd}$$

So, $A \otimes_R B$ serves as a coproduct in R-Alg, category of commutative R-algebra.

Problem 12

Compute the following tensor products:

- $k[x] \otimes_k k[x]$.
- $k[x] \otimes_{k[x]} k[x]$ where both $k[x]$ are biewed as $k[x]$-algebras via the identity map.
- $k[x] \otimes_{k[x]} k[y]$, where $k[x]$ is a $k[x]$-algebra via the identity map and $k[y]$ is viewed as a $k[x]$-algebra via the map $k[x] \to k[y]$ sending $x \mapsto y^2$.
- $k[y] \otimes_{k[x]} k[y]$, where $k[y]$ is viewed as a $k[x]$-algebra via the map $k[x] \to k[y]$ sending $x \mapsto y^2$.

Solution:

1. For any k-algebra A, given any two k-algebra homomorphism $f, g : k[x] \to A$, since one must have $f(l) = l \cdot f(1) = l \cdot 1 \in A$ (so is $g(l) = l \cdot g(1) = l \cdot 1 \in A$) for all $l \in k$. Hence, knowing $f(x)$ and $g(x)$ uniquely determines the whole map f, g.

   For simplicity, the second map we'll use y instead of x, or $g : k[y] \to A$ to prevent confusion. Then, notice that given the polynomial ring $k[x, y]$, f and g uniquely determins a ring homomorphism $h : k[x, y] \to A$ by $h(x) = f(x)$ and $h(y) = g(y)$. Such map is well-defined, since A is commutative, and the fact that $f(l) = g(l)$ for all $l \in k$ (so for everything inside k, where $k[x]$ and $k[y]$ intersects, h is a well-defined map). Also, h is a ring homomorphism because A is commutative, while f, g are both ring homomorphism (so act on individual components of any polynomial in $k[x, y]$ makes sense).

   Now, if consider the canonical inclusion $\iota_x : k[x] \hookrightarrow k[x, y]$ and $\iota_y : k[y] \hookrightarrow k[x, y]$, one has the following:

$$f(p(x)) = f(p(x)) \cdot g(1) = h(p(x) \cdot 1) = h(\iota_x(p(x))) \tag{5.35}$$

$$g(q(y)) = f(1) \cdot g(q(y)) = h(1 \cdot q(y)) = h\big(\iota_y(q(y))\big) \tag{5.36}$$

   Hence, $f = h \circ \iota_x$ and $g = h \circ \iota_y$, so the following diagram commutes:

$$\begin{tikzcd} k[x] \arrow[r, "\iota_x"] \arrow[dr, "f"'] & k[x, y] \arrow[d, "h"] & k[y] \arrow[l, "\iota_y"'] \arrow[dl, "g"] \\ & A & \end{tikzcd}$$

   Also, notice the map h is unique, since if $h' : k[x, y] \to A$ as a k-algebra homomorphism also satisfies the above properties, then $h'(x) = h' \circ \iota_{x(x)} = f(x)$ and $h'(y) = h' \circ \iota_{y(y)} = g(y)$. This shows that $h'(x) = h(x)$ and $h'(y) = h(y)$, hence $h' = h$ (since they're the same on the generators, while must agree on k due to the property of k-algebra homomorphism). Hence, $k[x, y]$ with ι_x, ι_y is indeed a coproduct of $k[x], k[y]$ in the category of commutative k-algebra, showing that $k[x, y] \cong k[x] \otimes_k k[y]$ (here, if view y as x again, it's $k[x] \otimes_k k[x]$).

2. For this, we'll prove a following lemma:

   Lemma

   Given any commutative ring R, any R-module M satisfies $M \cong R \otimes_R M$. This in particular is true for commutative R-algebra also.

   Proof: For any element in $R \otimes_R M$ is a finite linear combination of $r \otimes m$, where $r \in R$ and $m \in M$. Using the property of tensor product, $r \otimes m = 1 \otimes (r \cdot m)$.

   Hence, define the R-module homomorphism $\varphi : M \to R \otimes_R M$ by $\varphi(m) = 1 \otimes m$ (can also be thought of as the bilinear projection map $\pi : R \times M \to R \otimes_R M$ by restricting the first entry to be 1). On the other hand, if consider the scalar multiplication map $s : R \times M \to M$ by $s(r, m) = r \cdot m$, s is also a bilinear map, hence it uniquely factors through the tensor product, say having R-module homomorphism $\overline{s} : R \otimes_R M \to M$ by $s = \overline{s} \circ \pi$ (so we have $\overline{s}(r \otimes m) = \overline{s} \circ \pi(r, m) = s(r, m) = r \cdot m$).

   Notice that φ and $\overline{s}$ are mutual inverse: Given any $r \in R$ and $m \in M$, the following are true:

$$\overline{s} \circ \varphi(m) = \overline{s}(1 \otimes m) = s(1, m) = 1 \cdot m = m \tag{5.37}$$

$$\varphi \circ \overline{s}(r \otimes m) = \varphi(r \cdot m) = 1 \otimes (r \cdot m) = r \otimes m \tag{5.38}$$

Hence, this shows that $M \cong R \otimes_R M$ as R-modules. $\square$

As a result of the lemma, since $k[x]$ is a $k[x]$-module (with natural left multiplication on itself), then $k[x] \otimes_{k[x]} k[x] \cong k[x]$ as $k[x]$-modules (via the map $\overline{s} : k[x] \otimes_{k[x]} k[x] \to k[x]$ that's defined as $f(x) \otimes g(x) \mapsto f(x) \cdot g(x)$).

However, notice that this is also multaplicative, since the following is true:

$$\begin{aligned} \overline{s}((f(x) \otimes g(x)) \cdot (h(x) \otimes k(x))) &= \overline{s}((f \cdot h)(x) \otimes (g \cdot k)(x)) \\ &= (f \cdot h \cdot g \cdot k)(x) = (f \cdot g)(x) \cdot (h \cdot k)(x) \\ &= \overline{s}(f(x) \otimes g(x)) \cdot \overline{s}(h(x) \otimes k(x)) \end{aligned} \tag{5.39}$$

Hence, the two are actually isomorphic as rings, which is isomorphic as $k[x]$-algebras.

So, $k[x] \otimes_{k[x]} k[x] \cong k[x]$.

3. Similar in this case, using the previous lemma we know $k[x] \otimes_{k[x]} k[y] \cong k[y]$ via the map $\overline{s}(f(x) \otimes g(y)) = f(x) \cdot g(y) = f(y^2)g(y)$ (Note: Since $k[y]$ is now a $k[x]$-algebra via the map $x \mapsto y^2$, so $f(x) \cdot g(y) = f(y^2)g(y)$). This is also true on the level of $k[x]$-algebra homomorphism, since the following is true:

$$\begin{aligned} \overline{s}((f(x) \otimes g(y)) \cdot (h(x) \otimes k(y))) &= \overline{s}((f \cdot h)(x) \otimes (g \cdot k)(y)) \\ &= (f \cdot h)(y^2)(g \cdot k)(y) = (f(y^2)g(y)) \cdot (h(y^2)k(y)) \\ &= \overline{s}(f(x) \otimes g(y)) \cdot \overline{s}(h(x) \otimes k(y)) \end{aligned} \tag{5.40}$$

So, $k[x] \otimes_{k[x]} k[y] \cong k[y]$ as $k[x]$-algebra also.

Now, notice that if we consider the ring homomorphism $\varphi : k[x, y] \to k[y]$ by $\varphi(f(x, y)) = f(y^2, y)$, this is surjective (since for any $f(y) \in k[y] \subset k[x, y]$, it has $\varphi(f) = f(y)$), so $k[y] \cong k[x, y]/\ker(\varphi)$. Which, given the polynomial $y^2 - x$, we have $\varphi(y^2 - x) = y^2 - y^2 = 0$, so $y^2 - x \in \ker(\varphi)$.

Also, if consider any $f(x, y) \in \ker(\varphi)$ (i.e. $f(y^2, y) = 0$), then $f(x, y) = x \cdot f_1(x, y) + f_2(y)$ for some $f_1(x, y) \in k[x, y]$ and $f_2(y) \in k[y]$ (which are gathering the terms with x, and the terms with y only, respectively). There are two cases here:

- If $f_1(x, y) \notin \ker(\varphi)$, then $\varphi(xf_1(x, y)) = y^2 \cdot f_1(y^2, y)$ has degree at least 2; in case for $\varphi(f) = f(y^2, y) = 0$, this indicates that $f_2(y)$ must have
- Else if $f_1(x, y) \in \ker(\varphi)$, then it's evident that $f_2(y) \in \ker(\varphi)$, showing that $\varphi(f_2) = f_2(y) = 0$. Which, by induction one can break $f_1(x, y)$ into the previous case, and hence $f_1(x, y)$ is a multiple of $(y^2 - x)$.

6 ND

Problem 13

Prove that left adjoint preserves colimit.

7 D

Problem 14

Give an example that tensor product is not left exact in general.

Solution: Consider the $\mathbb{Z}$-module homomorphism $m : \mathbb{Z} \hookrightarrow \mathbb{Z}$ by $m(a) = 2a$ for all $a \in \mathbb{Z}$ (where m is injective). Then, it since $\mathrm{im}(m) = 2\mathbb{Z}$, this indicates that a cokernel of m can be given by $\pi : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ via canonical projection. Which, it forms the short exact sequence $0 \to \mathbb{Z} \hookrightarrow \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \to 0$ (the second map is m, the third map is π).

However, if consider the tensor product of the above ones with $\mathbb{Z}$-module $\mathbb{Z}/2\mathbb{Z}$, then the form the map $1 \otimes m : \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}$ by $(1 \otimes m)(a \otimes b) = a \otimes (2b)$ for all $a \in \mathbb{Z}/2\mathbb{Z}$ and $b \in \mathbb{Z}$. Yet, notice that this map is a zero map, since $a \otimes (2b) = (2a) \otimes b = 0 \otimes b = 0$ (because $2a = 0 \in \mathbb{Z}/2\mathbb{Z}$, where a is arbitrary). This shows that an injective module homomorphism after tensoring with some modules (and the corresponding identity map), it doesn't necessarily preserve injectivity (in this case, it turns m an injective morphism to $1 \otimes m$, a zero map).

Since this tensor product with $\mathbb{Z}/2\mathbb{Z}$ doesn't necessarily preserve injectivity, it doesn't preserve the left side of short exact sequence (since $0 \to \mathbb{Z} \hookrightarrow \mathbb{Z}$ by m, gets send to $\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}$ a zero map, which is no longer injective). So, tensor with $\mathbb{Z}/2\mathbb{Z}$ is not left exact, which serves as a counterexample here.